Teaching Dossier

Alain Maubert Crotte

Teaching Dossier

Alain Maubert Crotte | University of Regina

1. Teaching Experience

I have maintained a continuous teaching record at the University of Regina since 2016, spanning the Faculty of Science, the Faculty of Media, Art, and Performance, and the Faculty of Arts.

My teaching spans foundational lectures and upper-level production courses, including: * **CS 100: Introduction to Computers** and **CS 215: Web and Database Programming** * **CS 476: Software Development Project** and **CS 408: Animation Software Design** * **CTCH / DES 312: Introduction to Computer Game and VR Design** * **CTCH 223 / CS 207: Building Interactive Gadgets** (Lab Instructor) * **MAP 102: Exploring Cultural Regina**

In **CTCH / DES 312**, I contributed to the development of the course structure, content, and rubrics, guiding students through the full production pipeline from concept development to implementation. In **MAP 102**, I designed and delivered an experiential learning course that brought students into cultural institutions across Regina to critically interact with the city's creative organizations.

2. Course Design and Curriculum Development

I design courses around the principle that students learn a medium by making something that behaves, and by encountering the consequences of their decisions.

- **Learning Outcomes and Alignment:** I structure weekly modules with clear, measurable outcomes. In production courses, this means short demonstrations followed by studio time, and frequent small experiments rather than a single high-stakes assignment.
- **Iterative Assessment:** I design assessment frameworks including project guidelines, rubrics, and process documentation requirements. Students are evaluated not just on the final artifact, but on their ability to articulate why a design or technical decision was made.
- **Accessibility and Inclusion:** Drawing from my doctoral research on inclusive control schemes, I build accessibility into course design from week one, encouraging students to critically examine who technologies serve and how interfaces

include or exclude.

3. Online and Synchronous Teaching

I am experienced in delivering university courses fully online.

During the pandemic, I delivered **CS 100**, **CS 408**, **CS 476**, **CS 210**, and **SPAN** courses fully online. My approach utilizes: * **Synchronous Delivery:** I deliver lectures and labs synchronously via **Zoom** to maintain real-time interaction and student engagement, rather than relying on pre-recorded asynchronous media. * **Learning Management Systems:** I use **UR Courses (Moodle)** extensively across all courses for content delivery, readings, discussion forums, assignment submission, and grading. * **Classroom Technology:** In lab-based courses (CTCH 223 / CS 207), I manage hardware setup and troubleshooting for Arduino-based physical computing, as well as software configuration for Unity and Maya.

4. Teaching Methods and Student Engagement

My teaching methods emphasize active engagement and applied learning. * **Live Coding and Demonstrations:** Rather than reading from slides, I demonstrate concepts live, allowing students to see the problem-solving process in real time and ask questions as issues arise. * **Physical Computing and Hardware Access:** When teaching with specialized hardware (VR headsets, Arduino kits), I form balanced teams where each student has a specific role (operating equipment, developing assets, documenting processes) to ensure equitable access. * **Experiential Learning:** In MAP 102, I moved the classroom into the city, requiring students to engage directly with cultural institutions and creative professionals.

5. Equity, Diversity, and Inclusion in Teaching

I incorporate equity, diversity, and inclusion principles into course design and delivery. * **Inclusive Research:** My doctoral work on left-handed inclusion in computer games directly informs my pedagogy. I teach students to recognize right-handed and able-bodied defaults in technology. * **Diverse Materials:** I select diverse course materials and examples, ensuring that representation, identity, and cross-cultural perspectives are integrated into technical and media courses. * **Anti-Oppressive Practices:** Through the Graduate Student Teaching Development Program, I have applied principles of anti-oppressive teaching and Indigenizing curriculum, advocating for flexible assessment methods that allow students to demonstrate learning creatively.

6. Teaching Development and Continuous Improvement

I continuously refine my teaching through student feedback and curriculum modernization. * **Responsive Support:** When students struggle with dense material, I provide supplementary technical documentation (e.g., Unity manuals) and extend deadlines when necessary to ensure comprehension over strict pacing. * **Curriculum Updates:** I actively update course materials to reflect emerging industry practices,

moving away from legacy software toward modern pipelines like Unity and current VR SDKs. * **Reflective Practice:** I review student evaluations to adjust pacing and delivery, shifting from dense lectures toward more interactive, applied problem-solving.

7. Future Teaching Development

My future teaching goals include: * Expanding the use of interactive and applied learning approaches across all course levels. * Further integrating my research in accessibility and human-computer interaction into the Creative Technologies curriculum. * Supporting interdisciplinary learning across the Faculty of Media, Art, and Performance and the Faculty of Science, bridging the gap between technical implementation and creative practice.